Music Generation II

AI in Music, Winter Term 2025/2026

29.01.2026

Exercise 1 [Generation with MusicLang, 4 points]

In this exercise, you will use the MusicLang model to generate 10 pieces of music (Use seeds 0 through 9 for consistency). You can either use the Google Colab notebook that the authors provide, or follow their instructions on the URL above to install and run the model locally.
Note: For inference, you can run the model locally even without a GPU.

Exercise 2 [Evaluation, 4 points]

In the lecture, you learned about various evaluation measures that allow you to objectively measure important features in compositions. Implement the following evaluation measures:

- Overall number of different pitches used
- Pitch range, defined as the distance between the highest and lowest pitch
- Number of musical instruments used in the track
- The first two features from slide 16 from the lecture "13. Music Generation III", namely the repeated rhythm patterns of four notes f_{26}, and the rhythmic range f_{16}, but in seconds instead of ticks (removing the division by 16)

You will find ten MIDI files on Moodle from the LAKH dataset, which was used to train the MusicLang model. Calculate the above mentioned evaluation measures for the LAKH pieces, as well as for your generated pieces from Exercise 1. To normalise for differences in track duration, calculate the measures for every 10-second window and then average across the windows to create scalar values.

Exercise 3 [Detection of Generated Pieces, 2 points]

Use the evaluation measures from Exercise 2 to train 10 Random Forest classifiers (with the default parameters in scikit-learn), to classify a piece as "real" or "generated". Randomly split your dataset into 10 tracks for training and 10 tracks for testing, and use a different, *balanced* split on every seed. Report the average test classification accuracy achieved by your classifiers.